

Finding the best model for the given data using Machine Learning - Regression

1. Multiple Linear regression:

Evaluation metric: R_Squared (r2_score)

r2_score = 0.9358

2. SVM (Support Vector Machine):

Evaluation metric: R_Squared (r2_score)

S.No	Hyper Parameter	linear	rbf	poly	sigmoid
1	C = 1	0.8774	-0.05732	-0.0508	-0.0575
2	C = 10	0.5380	-0.05580	0.0253	-0.0576
3	C = 100	-107.97	-0.03022	0.4656	-0.0587
4	C = 1000	Kernal not responding	0.1606	0.6403	-0.0707
5	C = 2000	Kernal not responding	0.2884	0.6717	-0.0845
6	C = 3000	Kernal not responding	0.3951	0.6910	-0.0989

3. Decision Tree

Evaluation metric: R_Squared (r2_score)

S.No	Criterion	Splitter	Max_features	r2_score
1	squared_error	best	None	0.9409
2	squared_error	best	sqrt	0.7347
3	squared_error	best	log2	-0.2407
4	squared_error	random	None	0.9192
5	squared_error	random	sqrt	0.4206
6	squared_error	random	log2	-0.5826
7	absolute_error	best	None	0.9546
8	absolute_error	best	sqrt	0.2575
9	absolute_error	best	log2	0.4351
10	absolute_error	random	None	0.8765
11	absolute_error	random	sqrt	0.5667
12	absolute_error	random	log2	-0.4472
13	poisson	best	None	0.9158
14	poisson	best	sqrt	0.6544
15	poisson	best	log2	0.1740
16	poisson	random	None	0.8619
17	poisson	random	sqrt	0.5147
18	poisson	random	log2	0.6714

4. Random Forest

Evaluation metric: R_Squared (r2_score)

S.No	Criterion	Max_features	n_estimators	r2_score
1	squared_error	1.0	10	0.9028
2	squared_error	1.0	50	0.9411
3	squared_error	1.0	100	0.9450
4	squared_error	sqrt	10	0.5271
5	squared_error	sqrt	50	0.6845
6	squared_error	sqrt	100	0.7592
7	squared_error	log2	10	0.5271
8	squared_error	log2	50	0.6845
9	squared_error	log2	100	0.7592
10	absolute_error	1.0	10	0.9281
11	absolute_error	1.0	50	0.9401
12	absolute_error	1.0	100	0.9459
13	absolute_error	sqrt	10	0.7210
14	absolute_error	sqrt	50	0.7221
15	absolute_error	sqrt	100	0.7869
16	absolute_error	log2	10	0.7210
17	absolute_error	log2	50	0.7221
18	absolute_error	log2	100	0.7869
19	poisson	1.0	10	0.9304
20	poisson	1.0	50	0.9463
21	poisson	1.0	100	0.9420
22	poisson	sqrt	10	0.7477
23	poisson	sqrt	50	0.7242
24	poisson	sqrt	100	0.7732
25	poisson	log2	10	0.7477
26	poisson	log2	50	0.7242
27	poisson	log2	100	0.7732